

SENTRIQ architecture overview

Applies to SENTRIQ 4.2. A tour of the components and how telemetry flows from an endpoint to a detection. For a new operator or engineer.

The telemetry path

An agent on each endpoint collects process, network and file telemetry and streams it to a collector. Collectors normalise and forward to the event store. The detection engine evaluates rules against the stored stream and raises alerts to the console. Policy and signatures flow the other way: the policy service distributes configuration to agents, and the relay distributes signature updates.

The design goal throughout is that a failure in one stage degrades gracefully rather than losing data. Collectors buffer when the store is slow. Agents buffer when a collector is unreachable. The console can be down entirely without affecting collection or detection — it is only the operator's window, not part of the data path, and this separation is why a console outage (SQ-4002) never means endpoints are unprotected even though it looks like a total failure. The relay fails safe by refusing unverified content. Each buffer has a limit, past which data is dropped rather than held forever, and those limits are what the capacity plan (D17) exists to keep you inside of.

Components

- **Agent** — the endpoint component; collection, local detection, response actions.
- **Collector** — three nodes; receive, normalise, forward telemetry.
- **Detection engine** — two nodes; rule evaluation and correlation.
- **Event store** — four-node cluster; 90-day telemetry retention.
- **Policy service** — distributes policy, exclusions, and certificates.
- **Console** — the web UI and API; the operator's window.
- **Relay** — distributes signature and agent-version updates.

Trust boundaries

Agents authenticate to collectors and the policy service with mutual TLS. The certificate an agent holds is its identity; an expired certificate means the agent cannot be trusted and is cut off (see D07). This is why certificate expiry presents

as an offline endpoint rather than an error — the system is refusing an identity it can no longer verify.

Where things go wrong

Most incidents map to one stage of the path: agents offline (collection), collector back-pressure (forwarding), shard unassignment (storage), rule failure (detection), console logout (the window). Knowing which stage an alert belongs to is most of the triage, which is why the error codes are namespaced by component.

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